



PROJECT REPORT

Smart Electricity Bill Management System

A Java-Based Application for Automated Electricity Bill Calculation

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Date:	18-09-2026

1. Introduction

The Electricity Bill Management System is a Java-based application developed to simplify the basic process of electricity bill calculation. The system takes customer details and meter readings as input and calculates the total electricity bill based on the units consumed.

This project was developed to apply core Java concepts such as classes, objects, constructors, methods, conditional statements, input handling, and modular programming in a practical application.

2. Problem Statement

Electricity bill calculation involves different unit rates, fixed charges, and taxes. Doing these calculations manually can result in errors. The purpose of this project is to provide a simple system that performs these calculations automatically and displays the bill clearly.

3. Objectives

- To maintain basic customer information.
- To calculate electricity consumption from meter readings.
- To calculate the bill using different unit slabs.
- To include fixed charges and tax.
- To display the final bill in a clear format.
- To implement Java OOP concepts in a practical project.

4. Scope

The current project is a console-based application mainly focused on electricity bill calculation. It can be further developed into a complete billing system by adding a database, graphical interface, user login and online payment features.

5. Functional Requirements

The system provides the following functions:

- Enter and display customer details.
- Enter previous and current meter readings.

- Calculate units consumed.
- Calculate electricity charges according to unit slabs.
- Add fixed charges and tax.
- Display the complete bill.
- Display payment status.
- Validate meter readings.

6. Non-Functional Requirements

- **Usability:** The system should be simple to operate.
- **Accuracy:** Bill calculations should follow the defined slab rates.
- **Reliability:** Invalid meter readings should be handled properly.
- **Maintainability:** Different functions are separated into different classes.

7. Technologies Used

- **Programming Language:** Java
- **Concepts:** Object-Oriented Programming
- **Input:** Java Scanner
- **IDE:** VS Code
- **Version Control:** Git and GitHub

8. System Architecture

The project follows a simple modular structure:

User Input → Main.java → Customer / Bill / BillCalculator / Payment → Final Bill

Each class has a specific responsibility, which makes the program easier to understand and modify.

9. Project Structure

The project contains five main Java classes:

- **Customer.java** – manages customer information.

- **BillCalculator.java** – calculates energy charges.
- **Bill.java** – calculates and displays the complete bill.
- **Payment.java** – manages payment status.
- **Main.java** – handles user input and connects all the modules.

10. Working of the System

First, the user enters the customer information such as customer ID, name, meter number and address.

Next, the previous and current meter readings are entered. The system calculates the units consumed using:

Units Consumed = Current Reading – Previous Reading

The units are then passed to the Bill Calculator class, which calculates the energy charges according to the defined slabs. Fixed charges and tax are added to obtain the final bill amount.

Finally, the system displays the customer details, bill details and payment status.

11. Example Calculation

Suppose:

- Previous Reading = 1000
- Current Reading = 1250

Therefore:

Units Consumed = 1250 – 1000 = 250 units

Description	Amount (₹)
First 100 units	300
Next 100 units	500
Next 50 units	350

Energy Charges	1150
Fixed Charges	100
Tax (5%)	57.50
Total Bill	1307.50

12. Validation and Error Handling

The system checks whether the current meter reading is greater than or equal to the previous reading.

If the current reading is smaller, the program displays an error message instead of calculating an incorrect bill.

13. Testing

Different test cases were used to check the working of the system, such as:

- Lower unit consumption.
- Higher unit consumption.
- Different payment statuses.
- Valid meter readings.
- Invalid meter readings.

The results were checked against the expected calculations.

14. Challenges Faced

One of the main challenges was dividing the program into multiple classes instead of keeping everything inside one file. Understanding how objects and constructors connect different classes also required some practice.

After testing the individual modules, the overall structure became easier to understand.

15. Learning Outcomes

Through this project, practical experience was gained in:

- Java Object-Oriented Programming.
- Creating and using classes and objects.
- Using constructors and methods.
- Taking input using Scanner.
- Applying conditional logic.
- Modularizing a Java application.
- Basic project organization using Git and GitHub.

16. Future Enhancements

The project can be improved further by adding:

- MySQL database connectivity.
- User login and authentication.
- Graphical user interface.
- Monthly bill history.
- Online payment functionality.
- Automatic bill generation.
- Admin dashboard.

17. Conclusion

The Electricity Bill Management System is a simple but practical Java project that demonstrates how programming concepts can be used to solve a real-world problem. Working on this project helped improve understanding of Java OOP, modular programming, validation and basic software development practices.